

Task	Description	Steps	Status	Dependencies
Project Setup	Repository and LaTeX Setup	Create GitHub repo; set up folder structure; upload IEEE template; create references.bib; add README and planning folders	Done	None
Project Planning	Planning Strategy Definition	Define project phases (setup, literature, analysis, writing, revision, finalisation); map tasks to timeline (12 Mar – 1 May)	Done	
	Gantt Chart Creation	Create Gantt chart showing weekly timeline, task durations, and overlaps; export as PDF/image	Done	
	Task Breakdown & Milestones	Break project into weekly deliverables; define clear milestones (topic selection, literature complete, draft complete, final draft)	Done	
	Tool Setup for Tracking	Set up Trello/Notion board with columns (Backlog, This Week, In Progress, Done); add all tasks as cards	Done	
	Assignment Requirements Review	Read brief carefully; extract marking criteria; note page limit, repo requirements, viva expectations, and submission rules	Done	None
Topic Selection	Theme 3 Topic Exploration	Review all 3 Theme 3 topic options; read 2–3 overview/review papers per topic; compare difficulty, methods, and biological complexity	Done	Assignment Requirements Review
	Final Topic Decision	Select final Theme 3 topic; define provisional title and broad project direction	Done	Theme 3 Topic Exploration
Literature Review	Literature Search	Collect 10–15+ relevant academic papers from Google Scholar, PubMed/PMC, Springer, or similar sources	Done	Final Topic Decision
	Paper Matrix Creation	Create literature matrix with citation, problem, dataset, method, findings, limitations, and relevance to project gap	Done	Literature Search
	Core Paper Shortlisting	Select 4–6 strongest papers for deep critical review and project framing	Done	Literature Search
Critical Analysis	Deep Review of Core Papers	Analyse claims, methods, strengths, assumptions, weaknesses, and limitations of core papers	Done	Core Paper Shortlisting
	Methodological Clustering	Group papers by themes such as validation, robustness, generalisation, dataset shift, class imbalance, or predictive modelling strategy	Done	Deep Review of Core Papers
	Gap Identification	Identify what is known, what is weak, and what is missing in current literature; define a precise technical gap	Done	Methodological Clustering
Research Framing	Research Question Development	Write one focused research question aligned with identified literature gap	Done	Gap Identification
	SMART Objectives	Define 3–4 clear, feasible, and measurable project objectives	Done	Research Question Development
Methodology Design	Proposed Technical Workflow	Design project pipeline including input data, preprocessing, modelling logic, validation strategy, and evaluation approach	Done	SMART Objectives
	Dataset and Feasibility Review	Identify suitable bioscience dataset sources from literature; evaluate access, quality, suitability, and practical feasibility	Done	Proposed Technical Workflow
	Evaluation Strategy	Define metrics such as AUC, F1, calibration, robustness, or external validation criteria depending on topic focus	Done	Proposed Technical Workflow
Ethics & Risk	Ethical Risk Analysis	Identify risks including bias, privacy, reproducibility, imbalance, overfitting, and lack of generalisation	Done	Dataset and Feasibility Review
	Mitigation Planning	Propose safeguards and mitigation strategies for each ethical or practical risk	Done	Ethical Risk Analysis
Writing	Introduction Drafting	Draft problem context, motivation, and significance of chosen topic	Done	Research Question Development
	Literature Review Drafting	Write critical literature review showing current field state, limitations, and gap	Done	Gap Identification
	Methodology Section Drafting	Write proposed methodology, dataset considerations, evaluation plan, and expected workflow	Done	Evaluation Strategy
	Ethics and Feasibility Drafting	Write ethics, practical constraints, and mitigation strategies section	Done	Mitigation Planning
	Non-Technical Summary	Draft a short accessible explanation of the project for a non-specialist audience	Done	Introduction Drafting
Planning & Evidence	Gantt Chart Creation	Create project Gantt chart covering 12 March 2026 to 1 May 2026; export to PDF/image for planning folder	Done	Final Topic Decision
	Research Evidence Documentation	Save literature notes, gap analysis, paper matrix, and methodology sketches in research-evidence folder	Done	Literature Search
	GitHub Commit Trail	Make regular commits showing planning, reading, drafting, and revision progress across project lifecycle	Done	Repository and LaTeX Setup
Full Draft Development	Full Portfolio Assembly	Combine all written sections into one complete IEEE-format LaTeX draft and compile PDF	Done	Introduction Drafting, Literature Review Drafting, Methodology Section Drafting, Ethics and Feasibility Drafting
	Citation and Reference Check	Ensure all claims are cited properly and BibTeX entries are complete and consistent	Done	Full Portfolio Assembly
Revision	Critical Editing	Improve analytical depth, tighten arguments, strengthen transitions, and reduce descriptive summary	Done	Full Portfolio Assembly
	Page Limit Compression	Edit content down to 6 IEEE pages without losing critical quality	Done	Critical Editing
	Formatting and Proofreading	Check IEEE style, headings, figures, captions, spacing, references, and grammar	Done	Page Limit Compression
Finalisation	Final PDF Preparation	Generate final submission-ready PDF and ensure file naming is correct	Done	Formatting and Proofreading
	Repository Final Check	Confirm repo contains final PDF, LaTeX source, planning artefacts, README, and research evidence	Done	Final PDF Preparation
	Viva Slide Preparation	Prepare 3–4 concise slides covering problem, gap, method, and ethics/impact	Done	Final PDF Preparation
	Submission Readiness	Check Blackboard submission version matches GitHub PDF and all links/files work correctly	Done	Repository Final Check, Viva Slide Preparation